

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: ITA0605

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

***CO3: Bayes Theorem – Concept Learning – Maximum Likelihood – Minimum Description Length Principle – Bayes Optimal Classifier – Gibbs Algorithm – Naïve Bayes Classifier – Bayesian Belief Network – EM Algorithm – Probability Learning – Sample Complexity – Finite and Infinite Hypothesis Spaces – Mistake Bound Model,
(BL 6)***

Assessment Tool Weightage: 20%

QUESTIONS: 15

Total Marks:25

Annexure A

Case Study

Code of Conduct:

*I **THAMIZHARASAN S (192424087)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

THAMIZHARASAN

Annexure A

Short Answer Test

1, A hospital aims to develop an intelligent system to diagnose diseases based on patient symptoms and medical history. Using concepts such as Bayes' Theorem, Naïve Bayes classifiers, and Bayesian Belief Networks, design a probabilistic model for diagnosis. Analyze how uncertainty is handled and how conditional dependencies between symptoms influence predictions. Discuss the role of probability learning and evaluate the effectiveness of your approach with respect to real-world constraints.

ANSWER:

The hospital wants to develop an intelligent system to diagnose diseases using patient symptoms and medical history. The system can be designed using Bayes' Theorem, Naïve Bayes, and Bayesian Belief Networks (BBN). These probabilistic methods are suitable because medical diagnosis involves uncertainty and incomplete information. The question specifically requires analysis of uncertainty, conditional dependencies, probability learning, and real-world effectiveness.

1. Probabilistic model using Bayes' Theorem

Let:

- (D) = disease
- (S) = observed symptoms and medical history

Bayes' Theorem is:

$$P(D|S) = \frac{P(S|D)P(D)}{P(S)}$$

where:

- $P(D)$ = prior probability of the disease
- $P(S|D)$ = probability of observing the symptoms when the patient has the disease
- $P(D|S)$ = posterior probability of the disease after observing the symptoms
- $P(S)$ = probability of observing the symptoms

The system calculates the posterior probability for every possible disease and identifies the disease with the highest probability.

For example, if a patient has fever, cough, and fatigue, the system can calculate:

$$P(\text{Disease1}|\text{fever,cough,fatigue})$$

$$P(\text{Disease2}|\text{fever,cough,fatigue})$$

$$P(\text{Disease3}|\text{fever,cough,fatigue})$$

The disease with the highest posterior probability becomes the most likely diagnosis.

2. Naïve Bayes Classifier

Naïve Bayes assumes that symptoms are conditionally independent given the disease.

If the patient has symptoms $S_1, S_2, \dots, S_n$, then:

$$P(D|S_1, S_2, \dots, S_n) \propto P(D) \prod_{i=1}^n P(S_i|D)$$

For three symptoms:

$$P(D|\text{fever, cough, fatigue}) \propto P(D)P(\text{fever}|D)P(\text{cough}|D)P(\text{fatigue}|D)$$

The same calculation is performed for all candidate diseases.

Example:

Suppose for a particular disease:

$$P(D)=0.3$$

$$P(\text{fever}|D)=0.8$$

$$P(\text{cough}|D)=0.7$$

$$P(\text{fatigue}|D)=0.6$$

Then the unnormalized probability is:

$$\begin{aligned} &0.3 \times 0.8 \times 0.7 \times 0.6 \\ &= 0.1008 \end{aligned}$$

This value is compared with the corresponding values for other diseases. After normalization, the probabilities represent the likelihood of each disease given the patient's evidence.

Advantage: Naïve Bayes is simple, fast, and requires relatively few parameters.

Limitation: Its assumption that symptoms are conditionally independent may not always be true in medical diagnosis.

3. Bayesian Belief Network

A Bayesian Belief Network is more suitable when symptoms and medical-history variables have dependencies.

A simplified network can be represented as:

Medical History → Disease → {Fever, Cough, Fatigue}

For example, smoking history may influence respiratory disease, while respiratory disease may influence cough and breathing difficulty.

The joint probability in a Bayesian network is represented as:

$$P(X_1, X_2, \dots, X_n) = \prod_i P(X_i | \text{Parents}(X_i))$$

Therefore, the network does not simply treat every symptom as independent. It uses the conditional relationships between variables.

4. Handling uncertainty

Uncertainty is unavoidable in medical diagnosis because the same symptom can occur in different diseases.

For example, fever may be associated with several diseases. Therefore, the system should not simply produce:

“Fever = Disease X.”

Instead, it produces probabilities such as:

$$P(D_1 | \text{Evidence}) = 0.65$$

$$P(D_2 | \text{Evidence}) = 0.25$$

$$P(D_3 | \text{Evidence}) = 0.10$$

This means Disease 1 is the most probable diagnosis, but the system still recognizes uncertainty.

When new information such as a laboratory test or additional symptom becomes available, the posterior probability can be updated:

$$P(D|E_{old}, E_{new})$$

Thus, Bayesian inference allows the diagnosis to change as new evidence is received.

5. Influence of conditional dependencies

Conditional dependencies are important because medical symptoms are often related.

For example, suppose a patient has:

- Fever
- Cough
- Breathing difficulty

These symptoms may not be independent. A particular disease may cause several of them simultaneously.

Naïve Bayes treats them as conditionally independent:

$$P(F, C, B|D) = P(F|D)P(C|D)P(B|D)$$

A Bayesian Belief Network can instead explicitly represent relationships between the variables.

For example:

$$Disease \rightarrow Respiratory\ Condition \rightarrow Cough$$

This can provide a more realistic representation of the diagnostic process.

Therefore, the way conditional dependencies are modeled can directly influence the accuracy of disease prediction.

6. Role of probability learning

The probabilities required by the system can be learned from historical patient records.

For example, medical data can be used to estimate:

$P(\text{Disease})$

$P(\text{fever}|\text{Disease})$

$P(\text{cough}|\text{Disease})$

$P(\text{fatigue}|\text{Disease})$

If a large number of patients with confirmed diagnoses are available, the system can learn more reliable probability estimates.

The basic learning process is:

Historical Patient Data → *Estimate Probabilities* → *Build Probabilistic Model* → *Predict New Patient*

As additional reliable data become available, the model can be updated.

7. Effectiveness in real-world conditions

The proposed system can be effective because it:

- Combines symptoms and medical history.
- Provides probabilities rather than only binary decisions.
- Handles incomplete and uncertain information.
- Updates predictions when new evidence becomes available.
- Uses dependencies between medical variables through BBN.
- Learns probability values from historical patient data.

However, there are several limitations.

Data quality: Incorrect or incomplete patient records can produce inaccurate predictions.

Limited training data: Rare diseases may have insufficient examples for reliable probability estimation.

Naïve Bayes assumption: Symptoms may not actually be independent.

Incorrect network structure: A poorly designed Bayesian network can lead to incorrect inference.

Medical uncertainty: Symptoms alone may not be sufficient for a definitive diagnosis.

Real-world deployment: The system should be used as a clinical decision-support tool, with final decisions made by qualified medical professionals.

Overall workflow

Patient Symptoms + Medical History



Probability Learning



Bayes / Naïve Bayes / Bayesian Belief Network



Posterior Probability of Each Disease



Most Probable Diagnosis

Conclusion

A probabilistic disease-diagnosis system using Bayes' Theorem, Naïve Bayes, and Bayesian Belief Networks is appropriate for the hospital case. Bayes' Theorem provides the mathematical basis for updating disease probabilities from patient evidence. Naïve Bayes provides a simple and efficient classification method, while a Bayesian Belief Network can represent conditional dependencies between disease, symptoms, and medical history. Probability learning allows these probabilities to be estimated from historical patient data. The system can therefore handle uncertainty and provide useful diagnostic support, although its effectiveness depends on the quality of medical data, accuracy of probability estimates, correct modeling of dependencies, and appropriate clinical validation.

2. An organization wants to build a spam detection system for emails using machine learning techniques. Using Naïve Bayes and Bayes Optimal Classifier concepts, design a classification system. Compare Maximum Likelihood estimation with Minimum Description Length (MDL) principle in feature selection. Analyze system performance, discuss trade-offs in model complexity, and evaluate how sample size and hypothesis space affect classification accuracy.

ANSWER:

The organization can develop a spam detection system using Naïve Bayes and Bayes Optimal Classifier. The email is represented using features such as suspicious words, links, attachments, sender information, and email length.

1. Naïve Bayes Classification

Using Bayes' theorem:

$$P(C|X) = \frac{P(X|C)P(C)}{P(X)}$$

where (C) represents Spam or Not Spam and (X) represents the email features.

Naïve Bayes assumes that features are conditionally independent:

$$P(C|X) \propto P(C) \prod_i P(x_i|C)$$

The classifier calculates the probability of both Spam and Not Spam and assigns the email to the class with the higher probability.

2. Bayes Optimal Classifier

The Bayes Optimal Classifier considers different possible hypotheses and selects the classification with the highest overall posterior probability. It can provide better predictions because it considers uncertainty among possible models, but it can be computationally expensive when the hypothesis space is large.

3. ML vs MDL

Maximum Likelihood (ML) selects parameters that maximize the probability of the observed training data:

$$\hat{\theta}_{ml} = \arg\max_{\theta} P(D|\theta)$$

Minimum Description Length (MDL) selects a model by minimizing:

$$L(\text{Model}) + L(\text{Data}|\text{Model})$$

Thus, ML mainly focuses on fitting the training data, while MDL considers both data fit and model complexity. MDL can help reduce overfitting by avoiding unnecessary features.

4. Sample Size and Hypothesis Space

A larger training dataset generally provides more reliable probability estimates and improves generalization. A small dataset may produce unreliable estimates and overfitting.

A very small hypothesis space can cause underfitting, while a very large hypothesis space, especially with limited training data, can cause overfitting.

Conclusion

Therefore, an effective spam detection system should combine Naïve Bayes for efficient classification, Bayes Optimal reasoning for handling hypothesis uncertainty, and ML/MDL for probability estimation and feature/model selection. The final accuracy depends on appropriate model complexity, sufficient training samples, and a suitable hypothesis space.

3. A telecommunications company aims to predict customer churn using historical usage data and service feedback. Using probabilistic models such as **Naïve Bayes**, **Bayesian inference**, and hypothesis space search, design a predictive system. Analyze how prior probabilities and feature dependencies influence churn prediction. Evaluate how uncertainty in customer behavior is handled and discuss the effectiveness of the model in improving customer retention strategies.

ANSWER:

A telecommunications company can develop a customer churn prediction system using Naïve Bayes, Bayesian inference, and hypothesis-space search. The system uses historical information such as call usage, data usage, complaints, billing details, and customer feedback to predict whether a customer is likely to leave.

1. Naïve Bayes Prediction

Let (C) represent Churn or No Churn, and (X) represent customer features.

Using Bayes' theorem:

$$P(C|X) = \frac{P(X|C)P(C)}{P(X)}$$

Naïve Bayes assumes the features are conditionally independent:

$$P(C|X) \propto P(C) \prod_i P(x_i|C)$$

The system calculates the probability of churn and no churn and predicts the class with the higher probability.

2. Prior Probabilities and Feature Dependencies

The prior probability represents the probability of churn before considering the customer's current features.

For example, if historically 20% of customers leave:

$$P(\text{Churn}) = 0.20$$

This prior probability influences the final prediction.

Feature dependencies can also affect prediction. For example, poor service feedback and frequent complaints may be related. Naïve Bayes assumes conditional independence, while Bayesian inference can account for dependencies between customer factors.

3. Hypothesis Space Search

The system can consider different hypotheses, such as:

- High usage + poor feedback → churn
- Low usage + high complaints → churn
- Good feedback + low complaints → no churn

Searching the hypothesis space helps identify a model that best explains historical customer behavior.

4. Handling Uncertainty and Retention

Customer behavior is uncertain, so the system should provide probabilities rather than only a yes/no prediction. For example:

$$P(\text{Churn}|X) = 0.80$$

indicates a high-risk customer.

The company can then target high-risk customers with offers, improved service, or personalized support.

Conclusion

Thus, combining Naïve Bayes, Bayesian inference, and hypothesis-space search provides a probabilistic approach to churn prediction. Prior probabilities and feature relationships influence predictions, while uncertainty is represented through probabilities. The predictions can help the company identify high-risk customers and take timely actions to improve customer retention.

4. A smart healthcare system is designed to monitor patients in real-time using wearable sensors. Using **Bayesian Belief Networks** and probability learning, develop a model to detect anomalies in vital signs. Analyze how conditional dependencies between physiological parameters are modeled and how uncertainty due to sensor noise is handled. Evaluate system performance and discuss challenges in real-time deployment.

ANSWER:

A smart healthcare system can use a Bayesian Belief Network (BBN) and probability learning to monitor wearable-sensor data and detect abnormal changes in vital signs. The system considers parameters such as heart rate, blood pressure, temperature, and oxygen level.

1. Bayesian Belief Network

A BBN represents relationships between physiological parameters. For example:

Patient Condition->{Heart Rate,Blood Pressure,Temperature,SpO2

The network calculates the probability of a patient's condition based on the observed sensor values.

2. Conditional Dependencies

Physiological parameters can be related to each other. For example, a change in patient condition may simultaneously affect heart rate and blood pressure.

The BBN represents these dependencies using conditional probabilities:

$$P(X_i | \text{Parents}(X_i))$$

Thus, the system can make more meaningful predictions than treating every sensor measurement as completely independent.

3. Probability Learning

Historical patient and sensor data are used to learn probabilities such as:

$$P(\text{Abnormal} | \text{High Heart Rate}) \text{ and } P(\text{Abnormal} | \text{High Heart Rate, Low SpO}_2)$$

When new sensor readings arrive, the probabilities are updated and the system determines whether an anomaly is likely.

4. Handling Sensor Noise and Uncertainty

Wearable sensors may produce inaccurate or noisy measurements. Bayesian reasoning handles this uncertainty by assigning probabilities rather than making an absolute decision.

For example:

$$P(\text{Anomaly} | \text{Sensor Data}) = 0.85$$

indicates a high probability of an abnormal condition. Multiple sensor readings can be combined to reduce dependence on a single noisy measurement.

5. Real-Time Performance and Challenges

The system can continuously receive sensor readings and update the patient's probability of an anomaly. However, real-time deployment faces challenges such as sensor noise, missing data, computational requirements, communication delays, and the need for reliable probability models.

Conclusion

A Bayesian Belief Network provides an effective approach for real-time healthcare monitoring because it models conditional dependencies between physiological parameters and handles uncertainty caused by sensor noise. Probability learning from historical data improves the model's predictions, while continuous Bayesian updating enables early detection of abnormal patient conditions.

5. A logistics company wants to optimize delivery routes under uncertain traffic and weather conditions. Using **Bayesian inference** and probabilistic reasoning, design a system that predicts optimal routes dynamically. Analyze how prior knowledge and real-time data updates influence decision-making. Evaluate computational complexity and discuss how uncertainty impacts route optimization efficiency.

ANSWER:

A logistics company can use Bayesian inference and probabilistic reasoning to optimize delivery routes when traffic and weather conditions are uncertain. The system combines prior knowledge with real-time information to select the most suitable route.

1. Bayesian Route Prediction

For each possible route (R), the system can calculate its probability of being optimal based on traffic and weather conditions:

$$P(R|E) = P(E)P(E|R)P(R)$$

where (E) represents the current traffic and weather information.

The route with the highest posterior probability can be selected.

2. Role of Prior Knowledge

Historical information about traffic patterns and weather conditions provides the prior probability.

For example, if a particular road normally experiences heavy traffic during peak hours, the system initially assigns a lower probability to that route being optimal.

When real-time traffic information is received, the system updates the probability.

Prior Knowledge → Real – Time Data → Updated Probability

3. Dynamic Decision-Making

The system continuously receives information such as:

- Current traffic congestion
- Accidents or roadblocks
- Weather conditions
- Road travel time

If the originally selected route becomes congested, Bayesian inference updates the probabilities and another route can be selected.

Thus, the system can dynamically adapt instead of relying only on a fixed route.

4. Computational Complexity

Considering many routes, traffic conditions, and weather possibilities increases the number of probability calculations. Therefore, computational complexity can become high for large road networks.

A practical system should limit calculations to relevant routes and update only when significant changes occur.

5. Effect of Uncertainty

Uncertain traffic and weather conditions can make the predicted travel time inaccurate. Bayesian reasoning handles this uncertainty by assigning probabilities to possible conditions rather than assuming that a single condition is certain.

Therefore, the system can select a route that provides the best expected outcome under the current uncertainty.

Conclusion

Bayesian inference provides an effective method for dynamic route optimization because it combines historical knowledge with real-time traffic and weather data. The probabilities are continuously updated as new information becomes available. Although uncertainty and

computational complexity can affect performance, probabilistic reasoning allows the logistics company to make more flexible and informed routing decisions.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	20	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	15	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand